

When to Trust Simulated Data: Conformal Prediction for Selective Synthetic Data Integration

Sakir Furkan Yondem

SAKIRFURKAN.YOENDEM@TH-NUERNBERG.DE

Technische Hochschule Nürnberg, Keßlerplatz 12, 90489 Nürnberg, Germany

Seyda Yoncaci

SEYDAYONCACI@HALIC.EDU.TR

Haliç University, Faculty of Engineering, Department of Software Engineering, Istanbul, Türkiye

Gizem Karagoz

G.KARAGOZ@TUE.NL

Eindhoven University of Technology, Eindhoven, The Netherlands

Benedikt Schlereth-Groh

BENEDIKT.SCHLERETH-GROH@TH-NUERNBERG.DE

Technische Hochschule Nürnberg, Keßlerplatz 12, 90489 Nürnberg, Germany

Engin Avci

ENGINAVCI@FIRAT.EDU.TR

Firat University, Faculty of Technology, Department of Software Engineering, Istanbul, Türkiye

Ramin Tavakoli Kolagari

RAMIN.TAVAKOLIKOLAGARI@TH-NUERNBERG.DE

Technische Hochschule Nürnberg, Keßlerplatz 12, 90489 Nürnberg, Germany

Fatima Rabia Yapicioglu

PHD.FATIMA.RABIA.YAPICIOGLU@LAMBORGHINI.COM

Automobili Lamborghini S.p.A. & University of Bologna, DISI, Italy

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Abstract

Integrating simulated data into object detection training is a scalable strategy for alleviating data scarcity, yet simulated samples may exhibit geometric inaccuracies and domain artifacts that can degrade model performance when naively combined with real data. Rather than treating all synthetic samples as equally beneficial, we frame synthetic data inclusion as a reliability-aware selection problem and propose a conformal prediction-based framework to address it. The key insight is that a well-behaved synthetic sample should produce consistent object localizations across detectors trained on real and mixed data; violations of this consistency are captured as a nonconformity score, calibrated on a held-out set drawn from both real and synthetic data to derive a principled acceptance threshold. We evaluate on two benchmark datasets spanning autonomous driving and underwater sonar imaging, demonstrating consistent improvements in precision and recall across multiple model architectures. Explainability analysis further indicates that conformal selection yields more spatially coherent model attention, providing interpretable evidence that filtering acts on genuine domain artifacts rather than spurious correlations.

Keywords: Object Detection, Synthetic Data Refinement, Conformal Prediction, Uncertainty Quantification

1. Introduction

Although synthetic data can address the scarcity of real data in object detection, the key challenge is determining whether the generated samples contribute to training. Nevertheless, many hybrid training pipelines treat simulated samples as uniformly beneficial. The practical motivation behind this approach is clear: collecting and annotating large-scale real-world images is expensive, time-consuming, and in many domains simply impractical (Alzubaidi et al., 2023). Data augmentation offers partial relief but provides limited benefits when complex geometric variations or domain-specific characteristics are involved (Bansal et al., 2022). Simulated environments fill this gap by enabling scalable data generation with precise annotations and flexible scene control (Richter et al., 2016). However, simulation introduces its own problems: synthetic images frequently exhibit noise, geometric inaccuracies, or rendering artifacts that diverge from real-world distributions (Nikolenko et al., 2021). When naively combined with real data, these discrepancies can bias the learning process and degrade detection performance. This raises a fundamental question:

To what extent can geometric inconsistency between simulated and real data identify synthetic samples that are beneficial for real-world performance?

We argue that the answer lies in how detectors disagree. A synthetic sample that is geometrically consistent with the real world should produce similar object localizations regardless of whether the detector was trained on real data alone or on a mixture of real and synthetic data. Conversely, a problematic sample that violates real-world visual or geometric regularities, such as implausible shadow patterns, is expected to push the two detectors apart. This localization divergence is not noise: it is a structured signal about the trustworthiness of the synthetic sample. This leads to a second question:

How does geometric inconsistency, measured through inter-model localization disagreement, guide the definition of a principled threshold for synthetic sample admission?

Conformal prediction (CP) provides the statistical machinery to make this precise (Vovk et al., 2005). Uncertainty quantification (UQ) methods seek to estimate the confidence or reliability of model predictions, but classical approaches such as softmax scores or Monte Carlo Dropout provide only heuristic estimates with no formal statistical backing (Wang et al., 2025). CP addresses this limitation by operating as a post-hoc calibration layer that requires no modification to the underlying model. Given a held-out calibration set, CP measures how nonconforming a new sample is relative to the calibration distribution and derives a risk-controlled threshold at a user-defined miscoverage level α , without assumptions on the data distribution or model architecture (Angelopoulos and Bates, 2021). This makes CP particularly attractive in data-centric settings, where the reliability of individual samples, rather than aggregate model performance, is the primary concern. We exploit this by treating inter-model localization disagreement as a nonconformity score, calibrated on a held-out calibration set drawn from both domains, to obtain a principled acceptance criterion that admits only those synthetic samples whose detection behavior is consistent with real-world expectations. However, controlling which samples enter training is only half the story. This raises a final question:

How does conformal sample selection affect the spatial and semantic alignment of model attention as assessed by explainability metrics?

To answer this, we complement our quantitative evaluation with a LayerCAM-based explainability analysis, examining how selective data inclusion shapes the spatial attention of the trained detector. The result is a framework that is not only statistically principled but also interpretable: we can show *why* selected samples lead to better detectors, and *where* in the image the evidence lies. We validate the framework on two benchmark datasets spanning autonomous driving and underwater sonar imaging, domains that differ sharply in sensor modality, data volume, and visual complexity, demonstrating consistent performance improvements across multiple model architectures.

The main contributions of this work are as follows:

- *Reliability-Aware Hybrid Dataset Construction via Conformal Selection.* We reframe synthetic data inclusion as a selection problem and propose a conformal-calibrated filtering mechanism that excludes synthetic samples with high inter-model geometric disagreement, rather than treating all simulated data as uniformly beneficial.
- *Conformal-Calibrated Geometric Disagreement as a Nonconformity Measure.* We introduce inter-model IoU disagreement as a nonconformity score and compute a reproducible, data-dependent acceptance threshold using a held-out calibration set containing real and synthetic samples. Because this mixed calibration set is not exchangeable with the synthetic-only candidate set, the threshold does not provide the usual finite-sample distribution-free guarantee.
- *Empirical Validation Across Domains and Architectures.* We demonstrate consistent and transferable performance gains across two distinct data domains, autonomous driving and underwater sonar imaging, and multiple model architectures.
- *Explainability Analysis of Conformal Selection Effects.* We show through XAI-based analysis that conformal selection produces more spatially coherent and physically meaningful model attention, providing interpretable evidence that filtering acts on genuine domain artifacts rather than spurious correlations.

The remainder of this paper is organized as follows. We first review related work on synthetic data generation, sim-to-real domain gaps, and uncertainty-aware learning. We then present the proposed conformal selection framework, the experimental setup, and quantitative results, followed by the explainability analysis. We conclude with a summary of findings and directions for future research.

2. Related Work

2.1. Synthetic Data and the Sim-to-Real Gap

Simulated environments enable scalable data generation with precise annotations and flexible scene control, and have been widely adopted across domains including autonomous driving (Richter et al., 2016) and underwater sensing (Agrawal et al., 2025). However, models

trained on synthetic data frequently suffer from distributional shifts caused by rendering artifacts, geometric inconsistencies, and domain-specific biases (Kiefer et al., 2021; Nikolenko et al., 2021). Addressing this sim-to-real gap typically requires domain adaptation, domain randomization, or careful data curation — yet most hybrid training pipelines still treat all synthetic samples as equally beneficial, without assessing their individual reliability.

2.2. Uncertainty Quantification and Conformal Prediction

Uncertainty-aware learning is widely used in safety-critical perception tasks to evaluate the reliability of model predictions. In this context, conformal prediction has attracted increasing attention because it provides coverage guarantees that are distribution-free and finite-sample (Vovk et al., 2005; Angelopoulos and Bates, 2021). Recent work has extended CP to object detection, using it to bound localization risk at test time through margin-based conformity measures and conformal evaluation metrics (Mossina et al., 2025; Zouzou et al., 2025; Liu and Mohanty, 2025). Beyond test-time risk assessment, CP has also been used at training time to guide data selection in active learning, where samples are chosen for labeling based on conformal uncertainty (Kharazian et al., 2024). In contrast to these sample-labeling settings, we use CP to filter synthetic samples, employing inter-model localization disagreement as a conformity measure to decide which simulated data enter the training pipeline. Szabadvary (Szabadvary, 2026) identified *conformal blindness*, the failure of ill-suited conformity measures to detect distribution shifts, which is particularly relevant in the sim-to-real setting, where naive uncertainty metrics may overlook subtle simulation artifacts.

2.3. Explainability in Data-Centric Learning

XAI methods such as GradCAM (Selvaraju et al., 2016) and model-agnostic masking (Pet-siuk et al., 2020) have been used to verify whether models learn physically meaningful features or overfit to simulation artifacts. In data-centric settings, explainability serves not only as a diagnostic tool but as evidence that a selection mechanism acts on genuine domain characteristics rather than spurious correlations.

2.4. Research Gap

Although CP has been used for training-time data selection in active learning, it has not been applied to the reliability assessment of synthetic samples in sim-to-real settings. No prior work uses inter-model localization disagreement as a conformalized reliability signal for filtering synthetic samples before training. Our work closes this gap by introducing a disagreement-based nonconformity score that is less susceptible to conformal blindness, and by providing interpretable evidence of its effect through explainability analysis.

3. Methodology

In this section, we present our uncertainty-guided synthetic data refinement approach and its main components. Figure 1 summarizes the overall pipeline. First, we build two detectors: a reference detector $\mathcal{M}_r$, which is trained exclusively on real data, and a hybrid detector $\mathcal{M}_h$, which is trained on the unfiltered union of real and synthetic data. Next,

we quantify this discrepancy on synthetic samples using an intersection-over-union (IoU)-based nonconformity score, $s(x)$, which captures localization inconsistencies between the two models’ predictions. Finally, we statistically calibrate the selection threshold τ_α via split conformal prediction on a hybrid calibration set. We include only those synthetic samples that satisfy $s(x) \leq \tau_\alpha$ in the refined training set. The following subsections describe each step in detail.

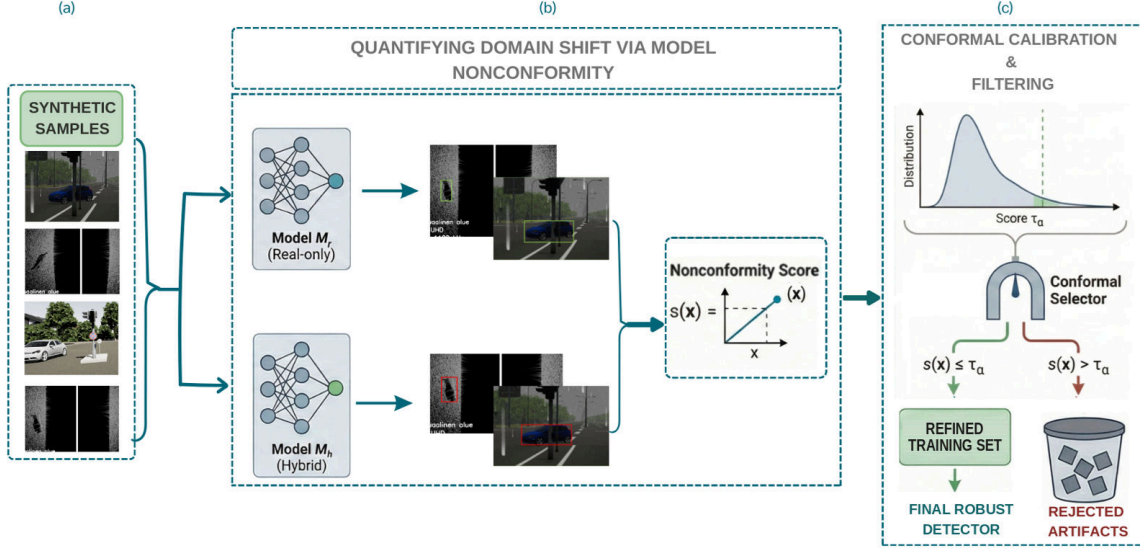


Figure 1: Uncertainty-guided synthetic data refinement.

3.1. Uncertainty-Guided Synthetic Data Refinement

Our approach views the sim-to-real gap as a sample-level reliability problem. Synthetic images may contain rendering artifacts, geometric inconsistencies, or domain-specific biases that can mislead the detector and cause overfitting to physically implausible cues. Our key assumption is that such problematic samples manifest as increased disagreement between detectors exposed to synthetic data to different extents. If a synthetic sample is consistent with the appearance and structural cues observed in real data, then $\mathcal{M}_r$ and $\mathcal{M}_h$ should localize objects similarly; in contrast, samples that distort object appearance or structure are expected to produce more divergent localizations. For example, in sonar imagery, a target is often observed together with an acoustic shadow whose relative position and extent depend on the viewing geometry. Synthetic artifacts that disrupt this target–shadow configuration can lead to unstable localizations, which may be reflected as increased disagreement between $\mathcal{M}_r$ and $\mathcal{M}_h$.

3.2. Quantifying Model Disagreement via IoU

Let $\mathcal{D}_{real}$ denote the labeled real-world dataset and $\mathcal{D}_{syn}$ the labeled synthetic pool. For each $x \in \mathcal{D}_{syn}$, we define an IoU-based nonconformity score $s(x)$ that quantifies localization disagreement between $\mathcal{M}_r$ and $\mathcal{M}_h$ based on their predicted bounding-box sets $\mathcal{B}_r(x)$ and

$\mathcal{B}_h(x)$. To quantify geometric nonconformity, we perform greedy one-to-one matching: for each $B_r \in \mathcal{B}_r(x)$, we select the still-unassigned $B_h \in \mathcal{B}_h(x)$ with the highest $\text{IoU}(B_r, B_h) = \frac{|B_r \cap B_h|}{|B_r \cup B_h|}$ (Everingham et al., 2010), and keep the match only if it is $\geq \tau_{\text{IoU}}$, so that each box is matched at most once. Following the PASCAL VOC (Everingham et al., 2010) and COCO AP50 standards (Lin et al., 2014), we set $\tau_{\text{IoU}} = 0.5$. This threshold enforces strict spatial agreement, filtering out accidental overlaps and ensuring that disagreement reflects genuine divergence in object perception. Let $n(x)$ be the number of matched pairs and their overlaps be $\{\text{IoU}_1, \dots, \text{IoU}_{n(x)}\}$. The mean overlap is:

$$\overline{\text{IoU}}(x) = \frac{1}{n(x)} \sum_{k=1}^{n(x)} \text{IoU}_k \quad (1)$$

Consequently, we define the nonconformity score $s(x)$ as:

$$s(x) = \begin{cases} 1 - \overline{\text{IoU}}(x), & n(x) > 0, \\ 1, & n(x) = 0 \end{cases} \quad (2)$$

where $n(x) = 0$ indicates that no prediction pair satisfies the matching criterion and we therefore assign the maximum nonconformity $s(x) = 1$ (see Figure 1(b)). Overall, larger values of $s(x)$ correspond to stronger localization mismatch between $\mathcal{M}_r$ and $\mathcal{M}_h$. To illustrate this effect, Figure 2 shows a foggy scene where $\mathcal{M}_h$ detects a few distant vehicles that $\mathcal{M}_r$ misses. For objects detected by both models, the predicted boxes are tightly aligned, indicating consistent shared geometry. Therefore, we compute $s(x)$ only over matched box pairs and do not penalize additional unmatched detections, as these may reflect useful additional findings rather than simulation artifacts. This approach minimizes unnecessary rejections and preserves synthetic variability, which can supplement limited real-world training data.



Figure 2: Example of IoU-based model agreement on a synthetic VKITTI2 image

3.3. Conformal Calibration of the Selection Threshold

Directly applying an arbitrary cutoff to the nonconformity score $s(x)$ provides no statistical control over which synthetic samples are included. To address this, we calibrate the selection threshold using *Split Conformal Prediction* (Figure 1(c)). We reserve an independent calibration set $\mathcal{D}_{cal} \subset (\mathcal{D}_{real} \cup \mathcal{D}_{syn})$ and compute calibration scores $s_i = s(x_i)$ for each

$x_i \in \mathcal{D}_{cal}$. Let $m = |\mathcal{D}_{cal}|$ and let $s_{(1)} \leq \dots \leq s_{(m)}$ denote the sorted calibration scores. Given a target miscoverage level $\alpha \in (0, 1)$, we set the conformal threshold to the $(1 - \alpha)$ empirical quantile with the standard finite-sample correction:

$$\tau_\alpha := s_{(\lceil (m+1)(1-\alpha) \rceil)} \quad (3)$$

Since τ_α corresponds to the $(1 - \alpha)$ percentile of calibration scores, increasing α selects a lower percentile and thus decreases τ_α . Under the acceptance rule $s(x) \leq \tau_\alpha$, a smaller τ_α makes the filter more selective. During the selection phase, a synthetic sample $x_{syn} \in \mathcal{D}_{syn}$ is accepted if

$$s(x_{syn}) \leq \tau_\alpha \quad (4)$$

Standard conformal validity requires exchangeability between calibration and test samples. Motivated by the need for robust calibration under distribution shift (Yu et al., 2022), we use a hybrid calibration set that mixes real and synthetic data. Since selection is applied only to synthetic candidates, the usual finite-sample coverage guarantee does not apply. We instead use the conformal quantile as a reproducible, data-dependent alternative to a hand-tuned threshold. Figure 3 illustrates the resulting partition: accepted samples lie mainly in the low-score region, whereas rejected samples concentrate at higher scores. Since this separation follows partly from the thresholding rule, the practical value of the method is assessed through downstream detection performance.

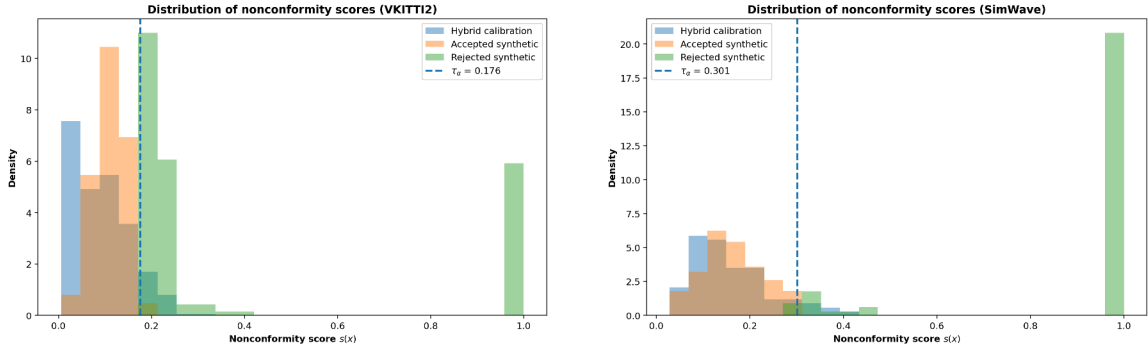


Figure 3: Nonconformity score distributions for the hybrid calibration set, accepted synthetic subset, and rejected synthetic subset across the VKITTI2 and SimWave.

The source code and implementation details are publicly available at <https://github.com/furkanyondemm/Trust-Simulated-Data>.

3.4. XAI-Driven Validation: The Pointing Game

We employ XAI to verify whether the performance gains from our hybrid training arise from attending to physically relevant regions or from spurious learning driven by simulation artifacts. Specifically, we utilize LayerCAM to generate explanation maps. We selected LayerCAM because it combines class activation maps from multiple intermediate layers—ranging from shallow to deep—weighted by their gradients, producing finer and more spatially precise visualizations than approaches relying solely on the last layer (Jiang

et al., 2021). Furthermore, since this procedure is applicable to both intermediate feature maps in CNNs and reshaped patch-token representations in transformer-based models (Chefer et al., 2021), it ensures a fair comparison across different architectures. To quantitatively validate the alignment of these explanations with ground truth objects, we adapt the Pointing Game (PG) evaluation protocol (Zhang et al., 2018) for our single-class object detection scenario. The PG measures the spatial selectiveness of attention maps independent of classification accuracy. For every predicted bounding box, we identify the maximum activation point within the LayerCAM heatmap:

- **Hit (TP):** If the maximum activation point falls inside any ground truth bounding box.
- **Miss (FP):** If the point falls outside all ground truth boxes (i.e., on the background).

The Pointing Game Accuracy (PG_{Acc}) is then computed as:

$$PG_{Acc} = \frac{Hits}{Hits + Misses} \quad (5)$$

A higher PG_{Acc} indicates that the model’s attention is consistently focused on the physical object rather than background noise or simulation artifacts.

4. Experimental Setup

4.1. Datasets and Modalities

We evaluate our method on two imaging modalities that differ sharply in data volume and visual complexity. One is acoustic side-scan sonar, which has limited real data. The other is optical on-road driving, represented by the large-scale KITTI benchmark.

4.1.1. SONAR DOMAIN: SIMWAVE AND AALTONEN

The first set of experiments was conducted using side-scan sonar imagery. To represent the real world, we used the Aaltonen dataset, which contains 331 real side-scan sonar images of the human class. To enable direct comparison with our previous work, we kept the same data split ($\mathcal{D}_{train}^{Aaltonen} = 145$, $\mathcal{D}_{test}^{Aaltonen} = 125$) (Yöndem et al., 2025). Similarly, for the synthetic domain, we used the same fixed SimWave pool of 206 images without modification or expansion (Yöndem et al., 2025). Keeping both the real training set and the synthetic pool fixed isolates the proposed conformal selection mechanism as the primary experimental variable within the hybrid training pipeline.

4.1.2. TERRESTRIAL DOMAIN: KITTI AND VIRTUAL KITTI 2

To evaluate the generalizability of our proposed selective approach beyond the sonar domain, we used a standard sim-to-real benchmark consisting of the KITTI (Geiger et al., 2012) and Virtual KITTI 2 (VKITTI2) (Cabon et al., 2020) datasets. We implemented the following strategies to ensure a controlled experimental environment and eliminate noise resulting from domain-specific discrepancies.

- **Class Focus:** Since the VKITTI2 dataset lacks "pedestrian" and "cyclist" categories, we strictly isolated our analysis to the "car" class, which is common to both domains.
- **Data Splitting:** Following the standard literature split to ensure comparability with existing benchmarks, the KITTI dataset was partitioned into 3,712 training images and 3,769 evaluation images (Chen et al., 2017).

4.2. Data Partitioning and Refinement

To improve data quality and reduce the risk of label noise biasing the evaluation metrics, we applied an additional inspection and filtering protocol to the datasets. As illustrated in Fig. 4, we identified cases in KITTI where a car annotation was present even though the vehicle was not discernible in the image (i.e., false-positive labels). On the Virtual KITTI 2 side, we also observed that annotations may be retained for objects that are effectively invisible due to heavy occlusion.



Figure 4: The following qualitative examples illustrate image-annotation mismatches (red boxes) that motivate the proposed filtering protocol: KITTI (top) and VKITTI2 (bottom).

To prevent annotation-image inconsistencies from distorting method comparisons, we enforced the filtering criteria summarized in Table 1, which is aligned with the "Moderate" definition in KITTI (Geiger et al., 2012).

Table 1: Comparative filtering criteria for KITTI and VKITTI2 datasets

Parameter	KITTI	VKITTI2
Min. BBox Height	≥ 25 px	≥ 25 px
Max. Occlusion	Partly occluded	$\geq 40\%$ occlusion
Max. Truncation	$\leq 30\%$	$\leq 30\%$
Empty-frame Check	Removed	Removed

After filtering, the refined terrestrial dataset consists of 2,402 training images and 3,093 evaluation images. Each image contains at least one valid vehicle annotation. In the hybrid

training setup, 1,650 images were randomly sampled from the VKITTI2 dataset to create a synthetic candidate pool. In the terrestrial setting, the calibration set, denoted by $\mathcal{D}_{cal}$, consists of 1,000 real images drawn from the KITTI evaluation split and 500 synthetic images from VKITTI2. These calibration images are used only for conformal calibration and are excluded from both training and final performance evaluation. This follows the guideline of using approximately $n \approx 1000$ calibration examples for statistical stability (Angelopoulos and Bates, 2021). The remaining 2,093 real images are reserved exclusively for evaluation. In the sonar domain, where real data is limited, the calibration set includes 61 real images from Aaltonen and 30 newly generated synthetic SSS images. These images are likewise used only for conformal calibration and are not included in training or final performance evaluation. Despite the smaller calibration set, the resulting threshold remains statistically meaningful (Angelopoulos and Bates, 2021).

4.3. Model Architectures

To evaluate the proposed conformal-calibrated synthetic selection framework, we examined the YOLOv8 (Jocher et al., 2023), YOLOv11 (Jocher and Qiu, 2024), and RF-DETR architectures. Since YOLOv8 and YOLOv11 were the baseline detectors in our previous work, using them in this study allows us to directly and comparably assess the improvement gained by the proposed selective hybrid training over a naive hybrid approach. Both models are CNN-based, one-stage detectors that are widely adopted in real-time applications. To examine whether the benefits of our method are architecture-specific, we included RF-DETR, which follows a more recent, distinct paradigm. Unlike the CNN-based YOLO models, RF-DETR (Robinson et al., 2025) uses a transformer-based design with self-attention mechanisms.

4.4. Conformal Selection Configuration

The selectivity of the mechanism in Eq. 3 is controlled by the miscoverage parameter α . Table 2 reports the number of accepted synthetic samples across $\alpha \in \{0.05, 0.10, 0.15\}$. As expected from Eq. 3, increasing α tightens the acceptance threshold and admits fewer samples; this effect is smooth and monotonic across all datasets and architectures, without abrupt jumps that would indicate an unstable selection regime. We adopt $\alpha = 0.10$ for all main experiments as a moderate operating point that filters unreliable samples without being overly restrictive.

Table 2: Number of selected synthetic images across different α values

α	VKITTI2		SimWave		
	YOLOv11	RF-DETR	YOLOv8	YOLOv11	RF-DETR
Unfiltered	1650	1650	206	206	206
0.05	1483	1460	145	126	153
0.10	1412	1392	121	110	130
0.15	1342	1323	104	98	118

5. Results and Discussion

5.1. Detection Performance on Sonar Domain

Table 3 compares the detection performance of different training configurations in the sonar scenario, using both the real Aaltonen and synthetic SimWave datasets. Overall, incorporating synthetic data improves performance. However, the results show that this improvement depends on more than just the inclusion of synthetic data; it also depends on the quality of the selected samples. The random-subset approach shows that merely reducing the amount of synthetic data is insufficient since randomly selected samples may include examples that are poorly transferable or not well aligned with the real domain. In contrast, CP-selected hybrid training provides more stable improvements and underscores the importance of selecting synthetic samples better aligned with the real sonar domain. This improvement is reflected in the precision-recall trade-off for YOLOv8 and YOLOv11 and is particularly evident in the recall for RF-DETR. In underwater search-and-rescue scenarios, missed targets pose a high operational risk and cost, so the increase in recall is a significant practical gain.

Table 3: Comparison of detection performance on the Aaltonen sonar dataset. Here, mAP50 denotes the mean average precision at an IoU threshold of 0.50, while mAP50-95 denotes the mean average precision averaged over IoU thresholds from 0.50 to 0.95.

Model	Dataset	Synt	Precision	Recall	mAP50	mAP50-95
YOLOv8	Aaltonen (Real-only)	–	0.937	0.808	0.901	0.561
	Hybrid (Naive)	206	0.963	0.823	0.935	0.631
	Hybrid (Random)	121	0.961	0.824	0.932	0.622
	Hybrid (CP-selected)	121	0.981	0.834	0.945	0.642
YOLOv11	Aaltonen (Real-only)	–	0.859	0.881	0.923	0.621
	Hybrid (Naive)	206	0.898	0.921	0.946	0.648
	Hybrid (Random)	110	0.911	0.792	0.923	0.620
	Hybrid (CP-selected)	110	0.921	0.935	0.948	0.632
RF-DETR	Aaltonen (Real-only)	–	0.975	0.936	0.972	0.652
	Hybrid (Naive)	206	0.975	0.944	0.955	0.609
	Hybrid (Random)	137	0.968	0.956	0.968	0.612
	Hybrid (CP-selected)	137	0.968	0.992	0.987	0.665

5.2. Detection Performance on KITTI

As shown in Table 4, the selection strategy significantly impacts performance, even when the same number of synthetic samples is used. For YOLOv11, the CP-based selection mechanism reduces the synthetic pool by about 14.4%, preserving mAP while improving precision and recall. Unlike the random-subset baseline, which uses the same number of samples, the CP-selected subset maintains a better performance balance, indicating that the observed gains are driven by sample reliability rather than subset size. Similarly, for

RF-DETR, CP-selected hybrid training provides the strongest overall performance balance, whereas random selection leads to a noticeable drop in performance. The relatively limited mAP gains on KITTI are consistent with expectations because the real training set in this large-scale benchmark already supports strong generalization. Conversely, the more pronounced improvements observed in the sonar domain suggest that conformal selection is particularly effective when real data is limited.

Table 4: Detection performance comparison on the KITTI validation set

Model	Dataset	Synt	Precision	Recall	mAP50	mAP50-95
YOLOv11	KITTI (Real-only)	–	0.915	0.934	0.969	0.891
	Hybrid (Naive)	1650	0.938	0.941	0.971	0.896
	Hybrid (Random)	1412	0.937	0.943	0.970	0.891
	Hybrid (CP-selected)	1412	0.941	0.948	0.972	0.896
RF-DETR	KITTI (Real-only)	–	0.962	0.943	0.974	0.856
	Hybrid (Naive)	1650	0.967	0.949	0.974	0.856
	Hybrid (Random)	1392	0.948	0.944	0.961	0.854
	Hybrid (CP-selected)	1392	0.973	0.952	0.974	0.858

6. Explainability and Alignment Evaluation

6.1. Visual Analysis of Saliency Maps

In Sections 5.1 and 5.2, we demonstrated that CP-selected hybrid training enhances detection metrics in both the sonar and KITTI scenarios. In this section, we use LayerCAM-based heatmaps to provide a qualitative analysis and understand which visual evidence the model relies on to achieve these gains.

6.1.1. SALIENCY MAP ANALYSIS IN THE SONAR DOMAIN

To qualitatively assess the effect of the proposed selection mechanism, we conducted a LayerCAM-based heatmap analysis of the YOLOv8, YOLOv11, and RF-DETR architectures. Figure 5 provides a comparative visualization of the discriminative regions emphasized by different training regimes for the same sonar inputs. For YOLOv8 and YOLOv11, the real-only and naive-hybrid regimes often produce fragmented activations over the target. Sometimes, the attention weights shift toward the head or upper torso; at other times, they shift toward the lower extremities. This makes it difficult to consistently cover the target’s entire geometry. In contrast, RF-DETR exhibits a more centralized focus, likely due to its transformer-based global attention mechanism. In most cases, activations concentrate around the upper torso. However, under the CP-selected hybrid regime, heatmaps across all three architectures display a more consistent and holistic pattern along the target. The continuous spread of activations along the head-torso-leg axis is consistent with the idea that the selected synthetic samples are a better training signal for learning the target’s geometric continuity.

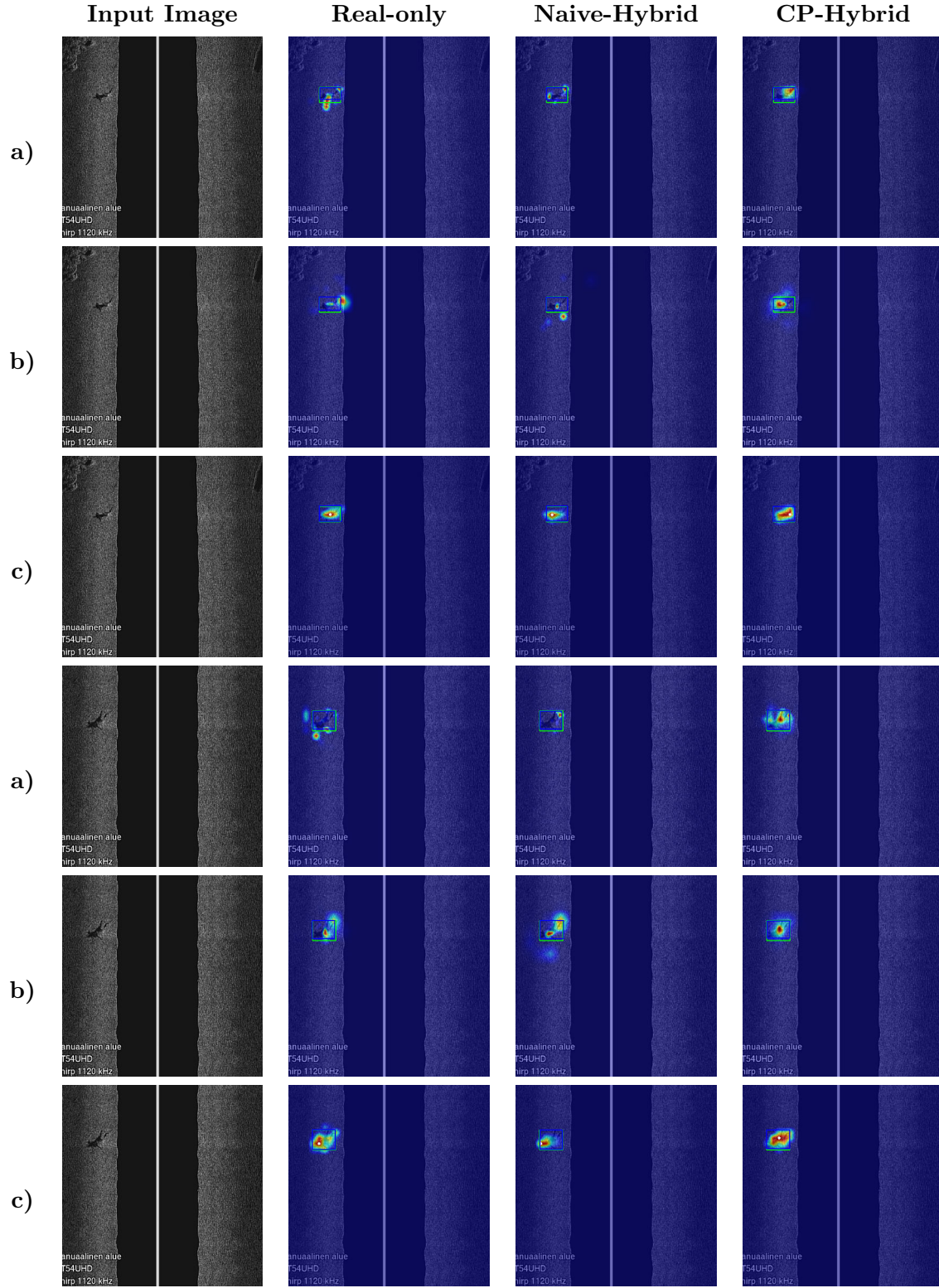


Figure 5: LayerCAM visualizations on sonar images. Panels: (a) YOLOv8, (b) YOLOv11, (c) RF-DETR. Columns compare Real-only, Naive-Hybrid, and CP-Hybrid regimes for the same input.

6.1.2. SALIENCY MAP ANALYSIS IN THE KITTI SCENARIO

Our LayerCAM analysis on KITTI allows us to qualitatively examine how the proposed method affects the separation of objects from the background in optical scenes. As Figure 6 shows, under the Real-Only and Naive-Hybrid setups, YOLOv11 allocates activation not only to target vehicles, but also to background regions, such as the road surface and surrounding textures. In contrast, YOLOv11 trained with the CP-selected hybrid regime concentrates activations more clearly on the vehicle body while suppressing the background response. For RF-DETR, a more "consolidated" attention pattern toward vehicles is observed across all regimes. However, under the real-only and naive-hybrid conditions, activations sometimes focus on specific parts of the vehicle. In contrast, in the CP-selected hybrid regime, the evidence spreads more broadly and consistently along the vehicle silhouette.

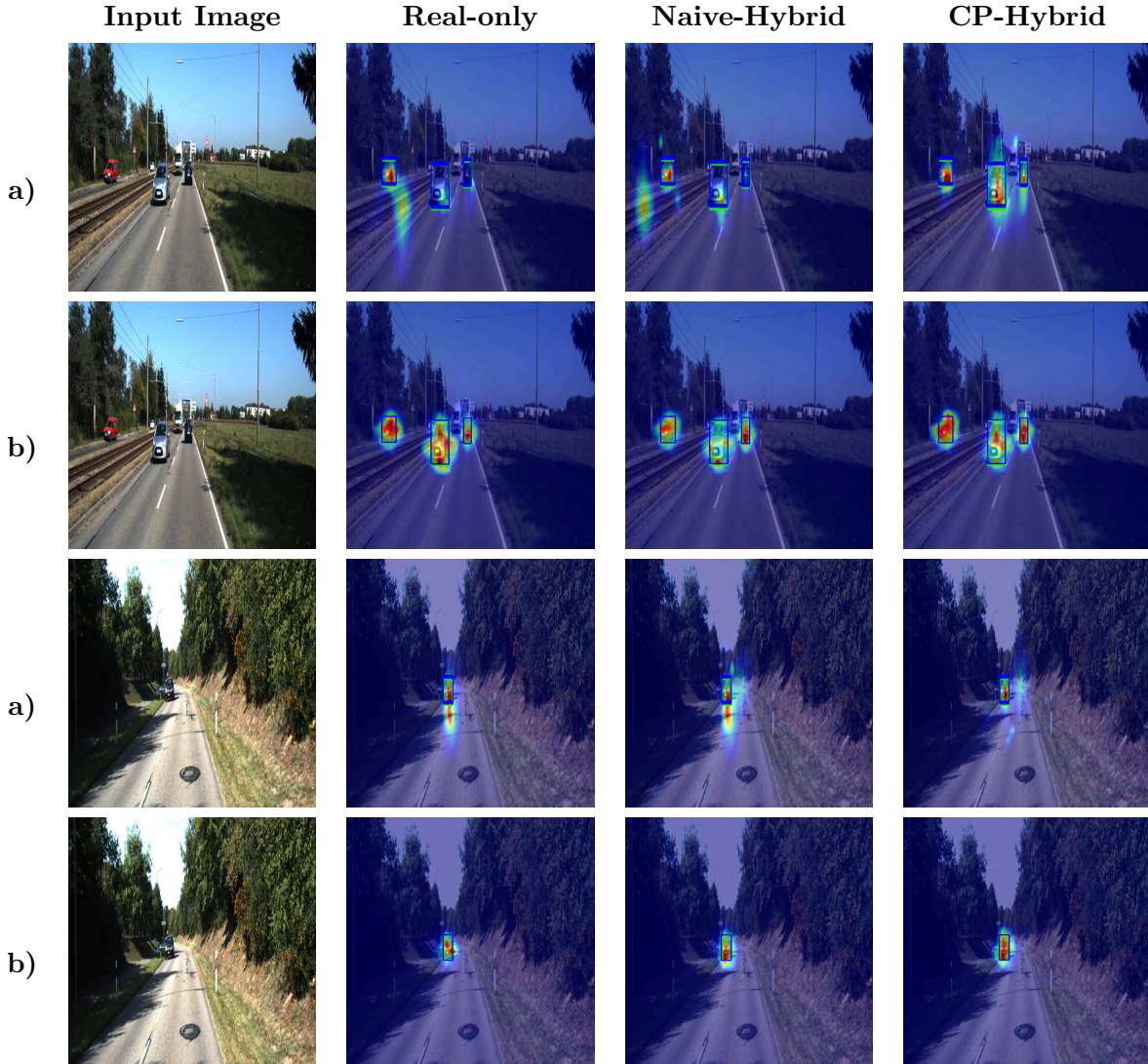


Figure 6: LayerCAM visualizations on KITTI samples. (a) YOLOv11, (b) RF-DETR.

6.2. Alignment Evaluation on Real-World Data

To assess the statistical significance of the improvements observed in the Pointing Game (PG), we applied McNemar’s test to the paired image-level hit/miss outcomes when comparing the CP-selected models with their real-only and naive-hybrid counterparts (McNemar, 1947). We conducted this analysis on both the Aaltonen and KITTI validation sets, and all comparisons showed statistically significant improvements ($p < 0.05$).

6.2.1. SONAR DOMAIN VALIDATION

As shown in Table 5, the results indicate that CP-based selection substantially improves the spatial alignment between the explanation maps and the target, particularly for low-resolution sonar images with pronounced speckle noise. For YOLOv11, CP-selected hybrid training achieves higher alignment accuracy than the real-only setup. This suggests that the model’s attention is more consistently concentrated on the target region. These results support the idea that the selection step filters out simulation-specific, mismatched samples, helping the model focus on physically meaningful, real-world cues (e.g., object–shadow structure). The transformer-based RF-DETR achieves the highest alignment accuracy between explanations and targets. This result demonstrates that, in addition to CNN-based detectors, selective hybrid training can also improve the quality of explanations for transformer-based models with different representation-learning assumptions.

Table 5: Alignment results via pointing game on the Aaltonen real validation set

Model	Val Dataset	Prediction	Hit	Miss	Accuracy
YOLOv8 (Real-only)	Aaltonen	96	49	47	0.511
YOLOv8 (Naive-Hybrid)	Aaltonen	96	65	31	0.677
YOLOv8 (CP-selected)	Aaltonen	99	71	28	0.717
YOLOv11 (Real-only)	Aaltonen	98	40	58	0.408
YOLOv11 (Naive-Hybrid)	Aaltonen	94	48	46	0.511
YOLOv11 (CP-selected)	Aaltonen	103	79	24	0.767
RF-DETR (Real-only)	Aaltonen	122	93	29	0.762
RF-DETR (Naive-Hybrid)	Aaltonen	119	107	12	0.899
RF-DETR (CP-selected)	Aaltonen	124	118	6	0.951

6.2.2. KITTI DOMAIN VALIDATION

Table 6 shows that the proposed method generalizes to the KITTI dataset. For RF-DETR, CP-selected training reduces the number of ”misses,” meaning peak activation points more often fall inside the ground-truth bounding boxes and improve explanation-target alignment. This suggests that selection mitigates the influence of the background and attention drift induced by domain mismatch, yielding more consistent evidence for the model’s decisions. Similarly, the higher PG scores achieved by the CP-selected YOLO and RF-DETR models

indicate that the proposed approach improves detection performance and produces more spatially consistent and target-aligned explanations.

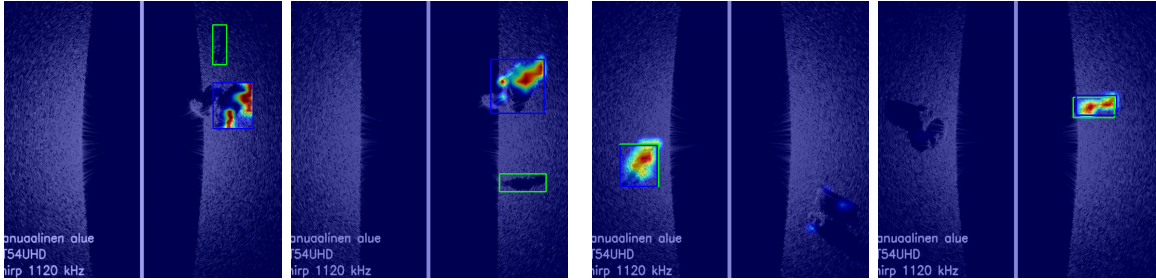
Table 6: Alignment Results via pointing game on the KITTI validation set

Model	Val Dataset	Prediction	Hit	Miss	Accuracy
YOLOv11 (Real-only)	KITTI	5461	4012	1449	0.735
YOLOv11 (Naive-Hybrid)	KITTI	5468	4130	1338	0.755
YOLOv11 (CP-selected)	KITTI	5420	4185	1235	0.772
RF-DETR (Real-only)	KITTI	5145	4511	634	0.877
RF-DETR (Naive-Hybrid)	KITTI	5175	4562	613	0.882
RF-DETR (CP-selected)	KITTI	5149	4707	442	0.914

6.3. Alignment Evaluation on Synthetic Data

To evaluate the effectiveness of the proposed selection mechanism, we performed a Pointing Game analysis on synthetic subsets that were labeled as "accepted" or "rejected" by the conformal-based selection process. For this analysis, we only use the real-only model. This reduces potential biases specific to quantitative evaluation of explanation alignment to assess whether the accepted subset exhibits more real-like attribution behavior. Additionally, we use LayerCAM analysis to determine which visual cues influence the acceptance and rejection decisions of the selection mechanism.

6.3.1. SALIENCY MAP ANALYSIS OF REJECTED SAMPLES



(a) Rejected samples

(b) Accepted samples

Figure 7: Comparison of synthetic samples rejected and accepted by the conformal mechanism based on their geometric consistency.

In the rejected example in Fig. 7 (left), these cues appear distorted or inconsistent. The model’s activation shifts toward background structures with similar shadow profiles (e.g., rocks or logs), indicating susceptibility to spurious cues. In contrast, the accepted sample in Fig. 7 (right), the body-shadow structure is more coherent, helping the model maintain attention on the target region and reducing activation leakage into the surrounding

clutter. This qualitative evidence suggests that conformal filtering acts as a quality gate by removing low-fidelity synthetic samples that could encourage shortcut learning and by promoting reliance on physically consistent visual cues.

6.3.2. SIMWAVE DATASET RESULTS

Table 7: Pointing-game alignment on SimWave: Subset indicates samples *Accepted* or *Rejected* by conformal selection

Model	Subset	Prediction	Hit	Miss	Accuracy
YOLOv8 (Real-only)	Rejected	30	7	21	0.233
	Accepted	121	77	44	0.636
YOLOv11 (Real-only)	Rejected	39	10	29	0.256
	Accepted	119	59	69	0.496
RF-DETR (Real-only)	Rejected	58	21	37	0.362
	Accepted	146	96	50	0.658

Table 7 shows the alignment results via PG for the Accepted/Rejected split of the SimWave synthetic data subset. The findings suggest that the CP-based filter can distinguish a clear misalignment between synthetic samples in terms of explainability. For example, the YOLOv8 model achieves low PG accuracy on the rejected subset but reaches a notably higher PG accuracy on the accepted subset. This indicates that the rejected samples may contain cues that divert the model’s attention away from the target, whereas the accepted samples enable the model to localize the target more consistently. Similar trends are observed for YOLOv11 and RF-DETR. The consistently higher PG scores of the accepted group indicate that the selection mechanism effectively identifies more “real-like” synthetic examples.

6.3.3. VKITTI2 DATASET RESULTS

Table 8: Pointing-game alignment on VKITTI2: Subset indicates samples Accepted or Rejected by conformal selection. Prediction is the number of evaluated predictions; Hit/Miss indicate whether the saliency peak falls inside/outside the ground-truth region; Accuracy = Hit/Prediction.

Model	Subset	Prediction	Hit	Miss	Accuracy
YOLOv11 (Real-only)	Rejected	532	395	137	0.743
	Accepted	3961	3125	836	0.789
RF-DETR (Real-only)	Rejected	752	588	164	0.781
	Accepted	3829	3338	491	0.872

Table 8 shows the alignment results via PG analysis for the Accepted and Rejected subsets of the VKITTI2 synthetic dataset. For both model architectures, higher PG accuracy in

the Accepted subset than in the Rejected subset indicates that the selection mechanism can identify synthetic samples that provide better explanation-target alignment in terrestrial imagery. The accuracy gap is approximately 4.6 percentage points for YOLOv11 and 9.1 percentage points for RF-DETR. Although the KITTI2 samples are cleaner and less noisy than the sonar images, these differences suggest that the proposed approach generalizes to highly noisy modalities and more conventional sim-to-real scenarios.

7. Conclusions and Future Work

We presented a conformal prediction-based framework for selective synthetic data integration in object detection, using inter-model localization disagreement as a nonconformity score to filter unreliable synthetic samples before training. Across two domains — data-scarce underwater sonar and large-scale autonomous driving — CP-selected hybrid training consistently improves precision and recall over naive hybrid baselines. Explainability analysis further shows that selected samples produce more spatially coherent model attention, concentrated on physically meaningful object cues rather than background or simulation artifacts. The clear alignment gap between accepted and rejected synthetic samples indicates that the conformal filter identifies more real-like examples, with consistent trends across both CNN-based and transformer-based architectures. We plan to extend the framework in two directions: investigating whether the conformal selection filter can serve as a quality gate for generative AI-produced samples by testing their statistical consistency with the target distribution, and adapting the IoU-based nonconformity score to segmentation tasks to broaden the approach to a wider class of vision problems.

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